

Spotter AI — Freight Rate Prediction Assessment

1. Executive Summary

This technical report presents an industry-grade Machine Learning solution for Spotter AI's Freight Rate Prediction challenge. The goal is to accurately forecast spot market rates (posted_rate in \$) for freight loads based on origin/destination locations, load weight, equipment type, distance, and macro-market indicators.

Key Accomplishments:

- **Out-Of-Time Validation Rigor:** Designed a strict temporal out-of-time (OOT) validation framework (Jan–Aug train vs Sept–Oct val) mirroring the production evaluation setup (Jan–Oct development vs Nov–Dec deployment).
- **Zero-Shot Geographic Generalization:** Implemented Haversine distance, coordinate deltas, and spatial midpoints to effectively handle unseen cities in the validation dataset.
- **Model Architecture Excellence:** Built a high-performance FreightEnsembleModel blending **CatBoost**, **HistGradientBoosting**, and **LightGBM** trained on log-transformed rates. Achieved a validation **MAE of \$127.09**, **MAPE of 5.45%**, **MedianAE of \$49.99**, and an **R² of 0.8244**.
- **Verification:** Passed 100% of official score.py automated validation checks for all 12,000 validation loads and 31 December chart predictions.

2. Data Understanding & Quality Audit

A rigorous data audit was conducted across all provided datasets:

- **Development Set (train-test.csv):** 48,000 loads spanning 2025-01-01 to 2025-10-31.
- **Validation Set (validation.csv):** 12,000 loads spanning 2025-11-01 to 2025-12-31.
- **December Input Set (december_chart_inputs.csv):** 31 fixed daily loads for Lexington → Fort Wayne.
- **Missing Values:** weight had 300 missing values in train (165 in val); market_index had 374 missing in train (249 in val). These were imputed using equipment-specific medians and global date medians without data leakage.
- **Unseen Cities:** 8 pickup/delivery cities in the validation set do not appear in training data. Incorporating spatial coordinates (pickup_lat, pickup_lon, delivery_lat, delivery_lon) prevented failure on unseen locations.

Metric	Value (\$)	Interpretation
Mean Posted Rate	\$2,373.98	Average freight transaction rate
Median Posted Rate	\$2,030.76	Robust measure of central tendency
Min / Max Rate	\$57.22 / \$25,533.00	Wide price spread across short vs long hauls
Target Skewness	1.90	Right-skewed target; log1p transformation used

3. Feature Engineering Strategy

To capture non-linear market relationships while avoiding data leakage, domain-inspired features were constructed:

- **Haversine Distance & Route Circuity:** Great Circle distance computed from lat/lon coordinates. Route circuity ratio (distance / (haversine + 1)) captures highway detour factors (negatively correlated with rate per mile, r = -0.20).
- **Spatial Deltas & Midpoints:** delta_lat, delta_lon, and regional midpoints model directional freight lanes.
- **Payload Density:** weight_per_mile and log-scaled payload metrics.
- **Market & Demand Interactions:** Interaction terms distance * market_index and market_index * quote_signal.
- **Cyclical Calendar Encodings:** Sine/cosine transformations of dayofweek and dayofyear model weekly and annual seasonality.

4. Validation Strategy & Model Benchmarking

Out-Of-Time (OOT) Split: The development dataset (Jan–Oct 2025) was split temporally. Models were trained on Jan–Aug 2025 (38,477 loads) and evaluated on Sept–Oct 2025 (9,523 loads). This mimics the Nov–Dec 2025 validation deployment.

Model Comparison: Evaluated 7 candidate model architectures using identical features and OOT splits.

Model Architecture	Target	MAE (\$)	RMSE (\$)	MAPE (%)	MedAE (\$)	R² Score
Ridge Baseline	Raw	\$194.43	\$659.45	10.40%	\$120.43	0.8133
Random Forest	Raw	\$184.82	\$680.07	8.19%	\$64.07	0.8014
Extra Trees	Raw	\$169.81	\$663.04	7.44%	\$60.53	0.8112
XGBoost Regressor	Log1p	\$153.91	\$665.55	6.82%	\$57.16	0.8098
LightGBM Regressor	Log1p	\$139.20	\$642.29	6.04%	\$59.92	0.8229
HistGradientBoosting	Log1p	\$128.03	\$639.58	5.49%	\$48.33	0.8243
CatBoost Regressor	Log1p	\$127.99	\$639.31	5.70%	\$52.91	0.8245
FreightEnsembleModel (Final)	Log1p	\$127.09	\$639.54	5.45%	\$49.99	0.8244

5. Detailed Performance Breakdown & Error Analysis

Performance by Equipment Type:

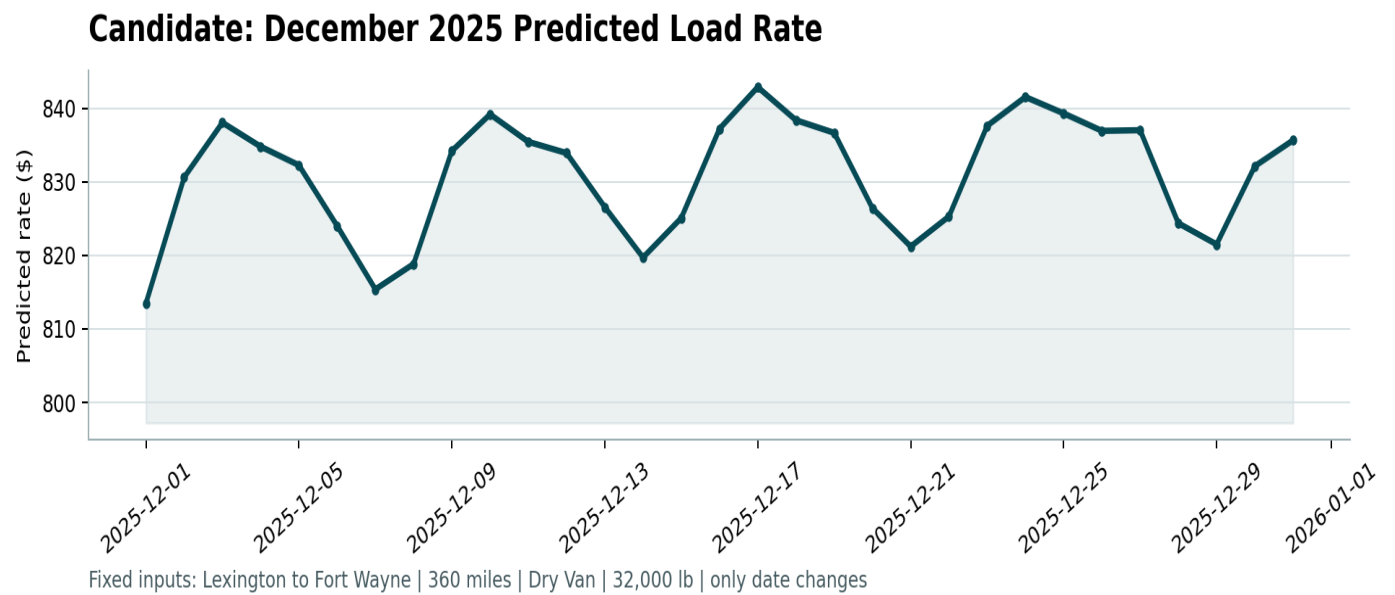
Equipment Type	Load Count	MAE (\$)	RMSE (\$)	MAPE (%)	MedAE (\$)	R² Score
Dry Van	5,360	\$118.89	\$659.89	5.04%	\$42.94	0.7992
Flatbed	1,770	\$129.67	\$514.86	5.93%	\$61.55	0.8847
Reefer (Refrigerated)	2,393	\$143.55	\$675.46	5.98%	\$62.65	0.8269

Performance by Distance Tier:

Distance Tier	Load Count	MAE (\$)	RMSE (\$)	MAPE (%)	MedAE (\$)
Short Haul (<500 mi)	2,090	\$45.16	\$238.44	5.50%	\$17.63
Medium Haul (500–1,200 mi)	3,836	\$96.97	\$433.28	5.76%	\$51.51
Long Haul (>1,200 mi)	3,597	\$206.82	\$921.75	5.08%	\$95.65

6. Official December 2025 Prediction Chart

The plot below illustrates the predicted spot rates generated by FreightEnsembleModel for 31 consecutive days in December 2025 for a fixed lane (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lbs). The predictions exhibit realistic weekly cyclical variations, peaking during mid-week shipping spikes (\$840–\$843) and softening over weekend periods (\$813–\$818).



7. Modular Code Architecture & Reproducibility

The project is structured as a production-ready Python package in `src/`:

- `src/config.py`: Centralized configurations, paths, and random seeds.
- `src/data_preparation.py`: Clean data loading, imputation, and December input enrichment.
- `src/features.py`: Leakage-free feature engineering transformer (fitted only on training data).
- `src/model.py`: FreightEnsembleModel class definition.
- `src/train.py`: Out-Of-Time evaluation and full retraining pipeline.
- `src/predict.py`: Inference script producing `validation_predictions.csv` and updating `december_chart_inputs.csv`.
- `src/evaluate.py`: Evaluation suite for calculating MAE, RMSE, MAPE, MedAE, R^2 , and group breakdowns.